

ETHAN BASS

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EDUCATION

The University of Chicago

Chicago, IL

- Bachelors of Science in Chemistry – Completed June 2013
- Cumulative GPA: 3.5 / 4.0
- Graduated with general honors

EXPERIENCE

Cedar Concepts Corporation

Chicago, IL

Quality Control Chemist – October 2013 to present

- Perform chemical analyses of materials at all stages of production and interpret data.
- Maintain and operate analytical instruments to ISO 9001 requirements including HPLCs, spectrophotometers, pH meters and autotitrator.
- Research the development of new products and the improvement of existing products and processes.

Department of Molecular Genetics and Cell Biology

Chicago, IL

Research Assistant – December 2011 to August 2013

- Designed and engineered mutations in recombinant DNA from *H. sapiens* and *P. falciparum*.
- Maintained insect cell lines and expressed recombinant proteins using a baculovirus expression system.
- Prepared microbiological media, buffers, plates, and competent *E. coli* cells.

International Genetically Engineering Machines Competition

Chicago, IL

Team Leader – March to October 2013

- Design and carry out a strategy for the isolation of recombinant DNA for expression in *B. subtilis*.
- Plan and execute assays for the detection and quantification of keratinase activity.

LABORATORY TECHNIQUES

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| • Bacterial culture (of <i>E. coli</i> and <i>B. subtilis</i>) | • DNA isolation and cloning | • Insect cell culture and transfection |
| • Competent cell preparation | • Plasmid preparation | • Protein expression and purification |
| • Genetic transformation | • PCR | • SDS-PAGE |
| • Preparation of media and reagents | • Site-directed mutagenesis | • Western blotting |
| | • Gibson Assembly | • HPLC maintenance and operation |
| | • DNA sequence analysis | |

COMPUTER SKILLS

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|-----------------------------|----------|-------------------|---------------------|------------------------------|
| • Microsoft and Open Office | • Matlab | • Adobe Photoshop | • Adobe Illustrator | • Sequencher |
| • ApE | • ImageJ | • PHREEQC | • PyMol | • NCBI and Pub Med databases |

PUBLICATIONS

Bass, E., Shan, K. 2013. Detection of Tellurium in Presence of Extra Arsenic on Gold Electrodes Modified By Benzenedithiol Self-Assembled Monolayers. *Electroanalysis*. 25(2): 503-506.